

ASWATHY GOPALAKRISHNAN

AI Engineer - Data Scientist - ML Engineer

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[LinkedIn](#)

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SUMMARY

AI Engineer and Data Scientist with 4+ years of industry experience and a Master's degree in Artificial Intelligence Engineering. Experienced in building end-to-end AI solutions using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector databases, machine learning, and structured reasoning frameworks. Strong background in Python, data science, analytics, prompt engineering, AI evaluation, and knowledge retrieval systems. Passionate about applying data-driven approaches to solve real-world business challenges, and continuously explore emerging AI technologies.

EDUCATION

Masters in AI engineering, Jönköping University, Sweden

Graduation Year (2026)

Bachelors in Electronics & Communication Engineering, Calicut University, India

Graduation Year (2011)

SKILLS

Programming and Data Tools: Python, R, KNIME, Tableau, Power BI, Advanced Excel, SQL, MySQL, SQL Server, NoSQL MongoDB, Vector Database (Qdrant, Chroma), Roboflow, GIT, Jira, Confluence.

Techniques: Data Analysis, Statistical Analysis, Exploratory Data Analysis (EDA), Data Science, Machine learning, Predictive Modeling, Deep Learning, Reinforcement Learning, Natural Language Processing (NLP), Large Language Models (LLM), Generative AI, retrieval Augmented Generation (RAG), Prompt Engineering, Explainable AI.

Soft Skills: Analytical thinking, Problem-Solving, Team collaboration, Adaptability, Research skills.

Languages: English (Professional), Swedish (L1 completed. L2 ongoing), Malayalam (Native) .

EXPERIENCE

Volvo Car Corporation, Gothenburg • AI Student Researcher - Master Thesis

(01/2026) - (06/2026)

- Designed and developed an end-to-end Generative AI application for automated root cause analysis of Android Automotive OS (AAOS) logs.
- Built Retrieval-Augmented Generation (RAG) pipelines leveraging hybrid retrieval and vector databases (Qdrant) to enrich LLM reasoning with domain knowledge.
- Implemented structured reasoning workflows and prompt engineering strategies to improve accuracy, explainability, and grounding of AI-generated analyses.
- Developed evaluation methodologies to assess evidence grounding, reasoning quality, hallucination rates, and diagnostic accuracy.
- Explored Agentic AI workflows using LlamaIndex-based tool-calling agents for autonomous investigation and synthesis.
- Conducted experiments on industrial datasets from Volvo Cars and public Android datasets, demonstrating improved performance through RAG and structured reasoning.

Technologies: Python, LLMs, RAG, Qdrant, LlamaIndex, Ollama, NLP, Chain-of-Thought prompting

Jordbruksverket, Jonkoping • Machine Learning Data Engineer Intern

(10/2025) - (12/2025)

- Evaluated and implemented annotation tools (CVAT, Roboflow) for agricultural image datasets. Built high-quality annotated datasets to improve ML model performance.
- Supported data preprocessing, model training, and performance evaluation. Applied interpretability techniques to improve model transparency.
- Communicated insights and progress with stakeholders and cross-functional teams.

Technologies: *Python, Roboflow, CVAT, Machine Learning, Data Creation*

Realdini, Stockholm • AI Developer Intern

(06/2025)-(09/2025)

- Developed text-to-audio and text-to-video pipelines using deep learning techniques.
- Worked with open-source LLMs to accelerate AI-driven game development workflows.
- Collaborated with designers and developers to deliver production-ready solutions.

Technologies: *LLM, TTS, python*

Labmentix EdTech Pvt. Ltd, India (Remote) • Data Analyst Intern

(05/2025) - (09/2025)

- Conducted exploratory data analysis to extract insights from real-world datasets.
- Built interactive dashboards using Tableau and Power BI.
- Performed data cleaning and transformation using Python and SQL.

Technologies: *Python, SQL, Power BI, Tableau, Streamlit, Advanced Excel*

Infosys Limited, India • Senior System Engineer

(07/2011) - (06/2015)

- Designed and optimized SQL queries, stored procedures, and PL/SQL functions to improve data integrity.
- Developed and maintained enterprise web applications using python Django and ASP.NET MVC. Worked across front-end (JavaScript, HTML, CSS) and back-end systems.
- Resolved critical production issues in collaboration with QA and operations teams. Mentored new engineers and contributed to technical documentation.

Technologies: *Python, ASP .NET, SQL, HTML, CSS, Javascript*

ACADEMIC PROJECTS

- Data analysis and ML model project for logistic company (Python, Knime)
- ML Cost Prediction Model for Manufacturing (Regression, Python)
- RDF-Based Music Recommendation System (RDF, SPARQL, Python, Knowledge graph)
- Bias Analysis in Crime Dataset using AI Fairness 360 (Data Analysis, ML, Python)
- Slow Fashion Image Classification (Deep Learning, CNN, ResNet, TensorFlow, Python)

ADDITIONAL EXPERIENCE

Freelance Technical Writer

(03/2020 – 10/2023)

Authored technical articles with emphasis on clarity and accuracy, for global clients and cross-cultural teams.

OTHER INFORMATION

- Valid Swedish work permit
- References available upon request